

COL RIVER BAR BUOY 46029, USA

WMO: 992370

Lat: 46.143N

Lon: 124.485W

Elev: 0

StdP: 101.33

Time Zone: -8.00 (W08)

Period: 01-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	2.2	3.9	-4.7	2.5	3.4	-2.2	3.1	5.1	17.2	9.0	15.7	9.7	7.0	90	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.7	17.7	16.6	17.3	16.2	16.8	15.9	16.8	16.9	16.3	16.4	15.9	16.1	4.8	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.8	12.0	16.8	16.3	11.6	16.3	15.8	11.2	15.9	47.1	16.7	45.5	16.5	44.5	16.2	18.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	14.1	11.8	DB	0.3	18.5	1.7	0.7	-0.9	19.0	-1.9	19.4	-2.8	19.9	-4.1	20.4	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.3	8.8	8.2	8.5	9.2	11.1	13.2	14.8	15.0	14.5	12.9	10.4	8.7
	DBStd	3.03	1.93	2.02	1.57	1.47	1.52	1.26	1.28	1.12	1.23	1.56	1.85	2.15
	HDD10.0	254	46	55	51	31	7	0	0	0	0	1	16	48
	HDD18.3	2569	297	284	305	273	226	155	109	102	113	169	237	298
	CDD10.0	728	7	4	4	8	39	95	150	157	137	91	28	8
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.2	7.3	6.9	6.8	6.3	5.4	5.4	5.2	4.7	5.3	6.2	7.2	8.0
Precipitation	PrecAvg	1883	288	195	200	149	80	71	28	46	49	190	284	294
	PrecMax	2470	587	542	431	299	133	132	85	132	193	383	540	515
	PrecMin	1409	112	18	21	58	10	19	6	3	0	42	81	136
	PrecStd	351	111	124	92	66	32	30	19	39	46	89	134	105
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.1	12.7	14.6	16.5	16.9	18.2	18.0	18.3	17.2	15.0	14.2
		MCWB	10.3	9.0	10.0	11.6	14.4	15.3	16.7	17.1	17.9	13.0	12.9	14.2
	2%	DB	12.3	12.3	12.1	12.8	14.6	15.9	17.6	17.6	17.4	16.3	14.1	12.9
		MCWB	10.5	9.1	10.1	10.9	13.0	15.2	16.7	16.7	15.9	13.5	12.5	12.0
	5%	DB	11.7	11.2	11.5	11.8	13.8	15.4	17.1	17.1	16.9	15.6	13.4	12.0
		MCWB	10.4	9.1	9.9	10.3	12.4	14.8	16.4	16.3	15.6	14.0	12.2	11.1
	10%	DB	11.1	10.5	10.6	11.2	13.0	15.0	16.7	16.7	16.5	15.0	12.8	11.3
		MCWB	10.0	8.9	9.5	10.1	12.0	14.3	16.0	15.9	15.4	13.9	11.8	10.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	10.6	11.3	12.1	15.1	16.2	17.3	17.8	17.8	15.2	13.5	12.4
		MCDB	11.9	10.7	11.5	13.7	16.3	16.3	17.3	17.8	17.8	15.3	13.6	12.6
	2%	WB	11.2	10.1	10.5	11.2	13.8	15.7	16.7	16.9	16.4	14.6	13.0	11.3
		MCDB	11.4	10.3	10.8	12.0	14.8	15.7	16.7	17.0	16.5	14.7	13.1	11.5
	5%	WB	10.8	9.6	10.1	10.8	12.9	15.2	16.2	16.4	15.7	14.1	12.6	10.6
		MCDB	11.1	10.0	10.4	11.1	13.6	15.2	16.3	16.5	15.9	14.3	12.8	10.8
	10%	WB	10.2	9.2	9.6	10.2	12.4	14.3	15.8	16.0	15.3	13.6	12.0	10.0
		MCDB	10.7	9.7	10.1	10.6	13.1	14.6	15.9	16.2	15.6	14.0	12.3	10.4
Mean Daily Temperature Range	5% DB	MDBR	2.1	2.1	2.0	1.9	1.6	1.5	1.5	1.7	1.9	1.9	2.2	2.3
		MCDBR	2.2	2.3	2.5	2.4	2.1	2.0	1.7	2.1	2.2	2.1	2.3	2.6
		MCWBR	2.1	2.0	2.3	2.6	2.0	1.8	1.9	2.1	2.2	2.2	2.1	3.0
	5% WB	MCDBR	1.9	1.8	2.1	2.0	2.1	1.7	1.8	1.9	2.2	1.9	2.2	2.3
		MCWBR	2.2	2.2	2.2	2.2	1.8	1.9	1.7	2.0	2.2	2.4	2.4	3.0
Clear-Sky Solar Irradiance	taub	0.327	0.324	0.340	0.374	0.390	0.363	0.352	0.354	0.349	0.341	0.330	0.322	
	taud	2.510	2.552	2.523	2.392	2.371	2.478	2.536	2.536	2.547	2.552	2.523	2.500	
	Ebn at Noon	775	860	892	884	878	902	908	895	871	826	765	736	
	Edh at Noon	66	76	90	111	118	106	99	96	87	75	65	61	
All-Sky Solar Radiation	RadAvg	1.17	2.05	3.17	4.53	5.38	5.80	5.71	4.95	4.11	2.52	1.39	0.99	
	RadStd	0.14	0.22	0.31	0.35	0.45	0.37	0.50	0.44	0.31	0.25	0.16	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+18

Nomenclature: See separate page